

Project Overview

- Project Name: Raphaaha Ecommerce Website
- Project Type: Role-based B2C Ecommerce + Admin Operations Platform
- Architecture: React frontend + Node/Express backend + MongoDB
- Primary Business Scope: Product catalog, shopping cart, checkout, online/COD payments, order lifecycle, offers, and website CMS settings
- Additional Operations Scope: Inventory, sales analytics, tasks, complaints, subscribers, and marketing broadcast

Technology Stack

- Frontend: React, Vite, React Router, Redux Toolkit, TailwindCSS, Chart.js/Recharts
- Backend: Node.js, Express, Mongoose, JWT, Bcrypt, Node-Cron
- Database: MongoDB
- Payment: Razorpay + PayPal integration paths
- Media: Cloudinary uploads and hosted image assets
- Communication: Email services, SMS OTP, web push utilities
- Documents: PDF invoice support

Current System Scale (From Codebase)

- Backend route modules: 33
- Backend models: 25
- Frontend pages/components: extensive user + admin dashboards with role-guarded paths

Role Model

- Guest/Public
- Customer
- Admin
- Merchantise
- Delivery Boy
- Marketing

Role-wise Module Access Matrix

Module	Guest/Public	Customer	Admin	Merchantise	Delivery Boy	Marketing
Auth (Register/Login/Reset)	Yes	Yes	Yes	Yes	Yes	Yes
Product Browsing & Search	Yes	Yes	Yes	Yes	Yes	Yes
Cart & Checkout	Yes/Partial	Yes	Yes	Yes	Limited	Limited
Orders (Own)	No	Yes	Yes	Yes	Limited by ops	Limited by ops
Reviews & Ratings	View	Submit/View	Moderate/View	Moderate/View	No	No
Wishlist	No	Yes	Yes	Yes	No	No
Offers Showcase	Yes	Yes	Yes	Yes	Yes	Yes
Website CMS (Hero/About/Policy/Contact Settings)	View	View	Manage	Manage	No	Limited View
Admin Product Management	No	No	Full	Full	No	No
Admin Order Management	No	No	Full	Full	Operational access	Operational access
User Management	No	No	Full	Full (as configured)	No	No
Inventory Management	No	No	Yes	Yes	No	No
Sales/Revenue/Trend Analytics	No	No	Yes	Yes	No	No
Task Management	No	Limited personal	All	Primary	Limited	Limited
Campaign Tracker & Broadcast	No	No	Oversight	Limited	No	Full
Subscribers & Contact Messages	Submit	Submit/View own interactions	Full	Full	No	Full
Complaints	Submit	Submit/View	Handle	Handle	Limited	Limited
Meta Options / Catalog Config	No	No	Yes	Yes	No	No
Delivery Registration	Yes	N/A	Yes	Yes	Yes	No

Core Functional Modules

1. Authentication and User Account Module

- User registration, login, and JWT token issuance

- Google login flow support.
- Forgot/reset password flows for users and admin—side reset utility.
- OTP—based mobile verification (‘send—otp’, ‘verify—otp’ flow).
- Profile update and role—aware session handling in frontend.

2. Role and Access Control Module

- Backend middleware: ‘protect’, ‘admin’, ‘adminOrMerchantise’, and generic ‘roleCheck(...)’.
- Frontend protected routes enforce role—based admin page access.
- Roles include operational variants beyond standard admin/customer.

3. Product Catalog Module

- Public product listing and product detail pages.
- Category/collection—specific pages (‘CollectionPage’, exclusive drop pages).
- Admin/merchant product CRUD through admin product routes and UI.
- Meta options for configurable catalog attributes.

4. Cart and Wishlist Module

- Cart add/update/remove/fetch for shopping flow.
- Wishlist add/remove/list for authenticated users.
- Cart drawer and checkout components integrated with product pages.

5. Checkout and Order Module

- Checkout pipeline with address capture and order confirmation.
- Order placement and lifecycle management.
- Customer order history and order detail pages.
- Admin/ops order board with status updates and operational controls.

6. Payment Module

- Razorpay route integration for online payments.
- PayPal button component available in frontend.
- Payment webhook endpoint for server—side payment event handling.
- Payment status integration into order state progression.

7. Review and Feedback Module

- Product review submission flow, including order—linked review screens.
- Review management routes for retrieval and moderation.

8. Website CMS / Content Settings Module

- Hero content management.
- About content management.
- Contact settings management.
- Privacy policy content management.
- Public consumption of settings—driven content pages.

9. Offers and Promotion Module

- Offer creation, editing, listing, and showcase pages.
- Scheduled email/offer broadcasts through scheduler services.
- Offer lifecycle support across admin and storefront.

10. Campaign Tracking Module

- Marketing—only campaign CRUD endpoints.
- Public click tracking and pixel impression routes.
- Conversion tracking endpoint for campaign performance measurement.

11. Subscriber and Contact Module

- Newsletter/subscriber collection and subscriber list views.
- Contact form submission and admin inbox views.
- Marketing broadcast interface for targeted communication.

12. Complaint Management Module

- Complaint submission and processing routes

- Suitable for customer grievance capture and internal response workflows.

13. Inventory and Sales Intelligence Module

- Inventory route set and admin pages.
- Sales trend analytics dashboard.
- Revenue reporting module for business monitoring.

14. Task and Internal Workflow Module

- Task creation, assignment–by–email, status update, and delete routes.
- Daily scheduler marks unresolved tasks as ‘not–completed’ after configured cutoff.
- Supports internal team workflow for merchant/admin operations.

15. Collaboration and Exclusive Drops Module

- Collab settings and preview admin screens.
- Exclusive drop pages and detail pages for campaign–driven merchandise launches.

16. File Upload and Media Module

- Secure upload endpoints protected by auth + admin middleware.
- Cloudinary image upload and deletion support.
- Product and site media asset lifecycle handling.

17. Address Management Module

- User address CRUD endpoints and checkout address integration.
- Supports delivery information persistence for repeat ordering.

Frontend Functional Areas

A. Storefront (Public + Customer)

- Home, collections, product details, cart, checkout, order confirmation.
- Login/register/profile/update profile.
- Offers, exclusive drops, about/contact/privacy pages.
- Wishlist, order history, review submissions.

B. Admin Portal (Role–Gated)

- Dashboard home and role–aware sidebar sections.
- Product management and product editing.
- Order management board.
- User management and hierarchy pages.
- Inventory, sales trends, revenue report.
- Website settings (hero/about/contact/privacy).
- Offers management and campaign tracker.
- Subscriber/contact monitoring and email scheduler.
- Task management and complaint handling.

Key Backend Data Entities

- User
- Product
- Cart
- Checkout / Checkout1
- Order
- Payment
- Review
- Wishlist
- Offer
- Campaign
- Task
- Subscriber
- Contact / ContactSetting
- Complaint
- Hero
- About

- Policy
- MetaOption
- Collab

Integration Summary

- Cloudinary: media upload/delete for product/site assets.
- Razorpay: payment initiation and verification/webhook processing.
- PayPal (frontend component support): alternate payment UX.
- Email services: welcome mails, marketing/offer broadcasts, transactional communication.
- SMS utility: OTP verification support.

Operational Automation

- Task status auto-update scheduler (daily rule-based update).
- Scheduled promotional email trigger service.
- Health check endpoint and self-ping behavior for uptime on hosted platform.

Security and Access Notes

- JWT-based authenticated API access for protected modules.
- Role-based middleware and frontend route guards are both used.
- Certain settings routes are implemented without explicit middleware and should be reviewed for stricter production hardening.

Functional Summary

Raphaaa Ecommerce Website is a full-stack commerce platform combining customer shopping workflows with strong internal operational tooling. It supports multi-role administration ('admin', 'merchandise', 'marketing', 'delivery_boy') and includes not only product-order-payment workflows but also campaign tracking, inventory intelligence, content management, and internal task governance.

Document Metadata

- Document Type: Functional Documentation (Role-wise + Module-wise)
- Project: Raphaaa Ecommerce Website
- Prepared From: Current codebase snapshot in '/Projects/Raphaaa'
- Update Recommendation: Revise after any route, role, or module change